

AUTONOMOUS RC CAR WEEK 4 PRESENTATION

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OUR GOAL AS A TEAM

Create an autonomous vehicle that can navigate through an unstructured environment.

Overall Steps

Step 01

Can it run autonomously without using the remote, with sensors and a microcontroller board to which we can upload code?

Step 02

Can it detect and identify obstacles correctly?

Step 03

Can it apply that obstacle detection to develop a software that can then avoid the obstacles?

Step 04 (Bonus)

Can it create paths based on given targets?

Progress since our last in person meeting/obstacles encountered

- Managed to finally successfully 3D print and mount both parts we had previously designed and tried printing
- Some loose wires have been resoldered
- Connected to Stevens Wifi which was unsuccessful last time
- Started integrating AI interface into obstacle detection
- Biggest obstacle encountered today was getting wifi to work

Demo of Camera System

Next Steps:

- Use AI interface (YOLO) to control throttle along with the steering
- Use camera detection and AI interface for more reliable obstacle avoidance
- Start considering onboarding AI products to use, and also Lidar sensors.

THANK YOU

Any questions?